

Course: Wireless and Mobile Security (CY315)

Semester: Spring-2026

Project Overview

Modern wireless networks (5G/6G, IoT, Wi-Fi 6) face sophisticated threats at both the protocol and physical layers. Standard security often fails against intelligent jamming, cross-layer attacks, or resource-constrained environments.

Students are required to conduct a research-based project that either **critically evaluates** existing WMS solutions (Survey Track) or **optimizes/enhances** a specific security mechanism (Implementation Track). The goal is to move beyond basic concepts to address:

- **Robustness:** Protecting communication against active eavesdropping or interference.
- **Efficiency:** Optimizing security overhead for mobile/constrained devices.
- **Contemporary Solutions:** Applying Physical Layer Security (PLS) or advanced cryptography to modern wireless problems.

Knowledge Areas

- Physical Layer Security (Jamming, Wiretapping, Key Generation).
- Cryptographic Protocols for Mobile Networks.
- Wireless Standards Security (IEEE 802.11, 5G/6G, LoRaWAN).
- Attack Modeling (Man-in-the-middle, Denial of Service, Spoofing).

Team Formation

- Groups: **1–3 members** (4 allowed only by prior instructor approval).
- Contribution: All members must actively participate. Contribution will be verified during the final viva.

Project Tracks & Deliverables

Each group must choose **one** of the following tracks:

Track 1: Systematic Research Survey

- **Objective:** Critically evaluate the state-of-the-art in a specific WMS sub-domain.

- **Deliverables:** A formal research paper (8–12 pages) in LaTeX/ overleaf (IEEE format).
 - Original Taxonomy Diagram and Comparative Evaluation Table.

Track 2: Implementation & Optimization

- **Objective:** Implement a security mechanism from a research paper and “Optimize” its performance or security.
- **Deliverables:**
 - Functional code/simulation (Python, MATLAB, NS-3).
 - A technical report (6–10 pages) in LaTeX/ overleaf (IEEE format) showing comparative results (Original vs. Optimized).

Report Structure

1. **Abstract:** Summary of the security problem and findings.
2. **Introduction:** Background of the wireless technology and threat model.
3. **Literature Review:** Analysis of existing work.
4. **Methodology/ The Proposed Evaluation/Optimization:** Detailed logic of your specific contribution.
5. **Results & Discussion:** Performance graphs, security analysis, and critical evaluation.
6. **Conclusion & Future Work.**
7. **References:** BibTeX formatted citations.

Assessment Criteria

- **Originality of Analysis (30%):** Does the project show depth beyond basic textbook knowledge?
- **Technical Execution (30%):** Quality of the survey taxonomy or the simulation results.
- **Report Quality (20%):** Professionalism, LaTeX formatting, and clarity.

- **Viva & Presentation (20%):** Individual understanding and defense of the chosen approach.

Plagiarism Policy

- Similarity index must not exceed 19% (Turnitin).
- AI-generated reports are strictly prohibited and will result in zero marks.

Submission Deadline: Week 14

Format: PDF (from LaTeX source) + Presentation Slides + Source Code (if applicable).